CH - 5 Reaction Mechanism of d-block complex

SUZON-50

Thermodynamic stability and libility have no direct relations. A compound which is thermodynamically stable may be labile (দ্রুত)

Ex:

$$[Ni(CN)_4]^{2-} + 4\,{}^{14}CN^- \rightleftharpoons [Ni({}^{14}CN)_4]^{2-} + 4CN^-$$

On the other hand, $[Co(NH_3)_6]^{3+}$ is not stable but it is inert (নিষ্ক্রিয়). In acid soln, it

$$2[Co(NH_3)_6]^{3+} + 10H^+ + 18H_2O \longrightarrow 2[Co(H_2O)_6]^{2+} + 6NH_4^+ + 1/2\,O_2$$

(very slow)

F-22 F-21

$$[Co(NH_3)_5(CO_3)]^{+} + 2H_3O^+ \longrightarrow [Co(NH_3)_5(H_2O)]^{3+} + CO_2 + 2H_2O$$

Mechanism:

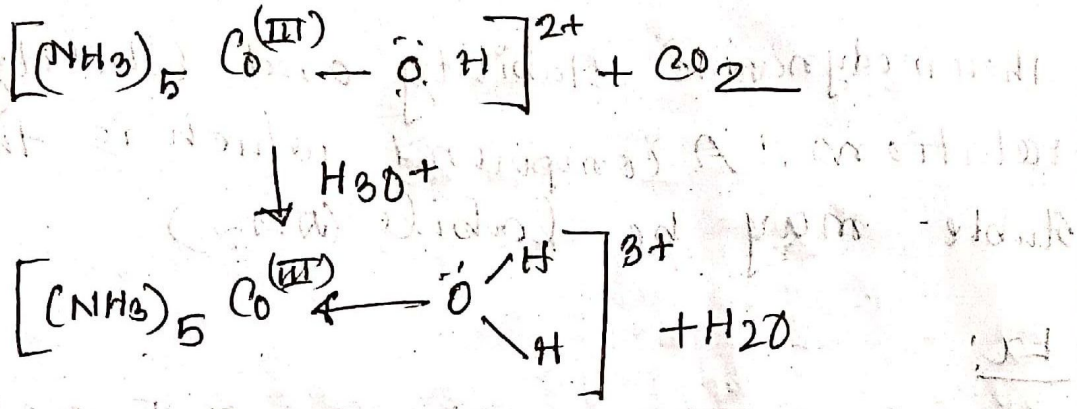
$$[(NH_3)_5 \overset{(III)}{Co} - \ddot{O} - CO_2]^+ + H_3O^+$$

$$\downarrow$$

$$[(NH_3)_5 \overset{(III)}{Co} - \ddot{O}(H^+)(CO_2)]^{2+} + H_2O$$

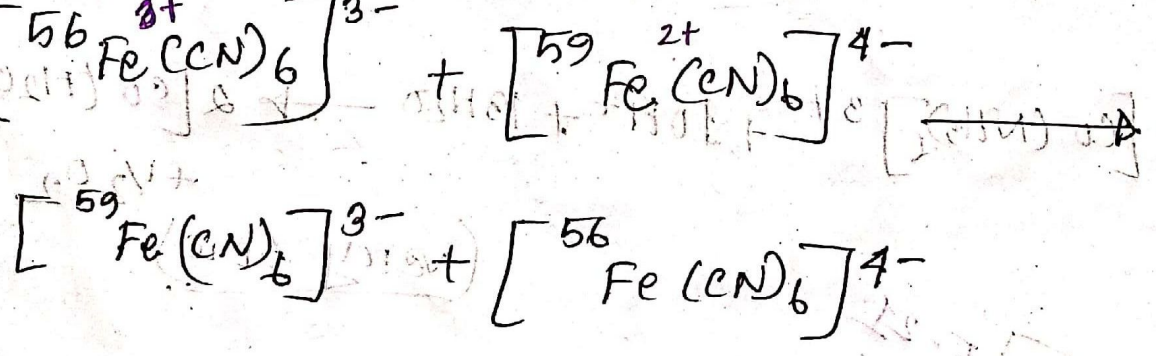
$$\downarrow$$

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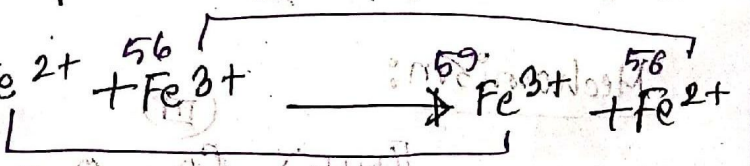


So, in this complex, "O" of H_2O is from CO_3^{2-} and is not replaced CO_3^{2-} by H_2O

[II]



This reaction is not substitution, it is a redox rxn. Actually, e^- transfers from $[^{59}Fe(CN)_6]^{4-}$ to $[^{56}Fe(CN)_6]^{3-}$



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Mechanism of Ligand Substitution:

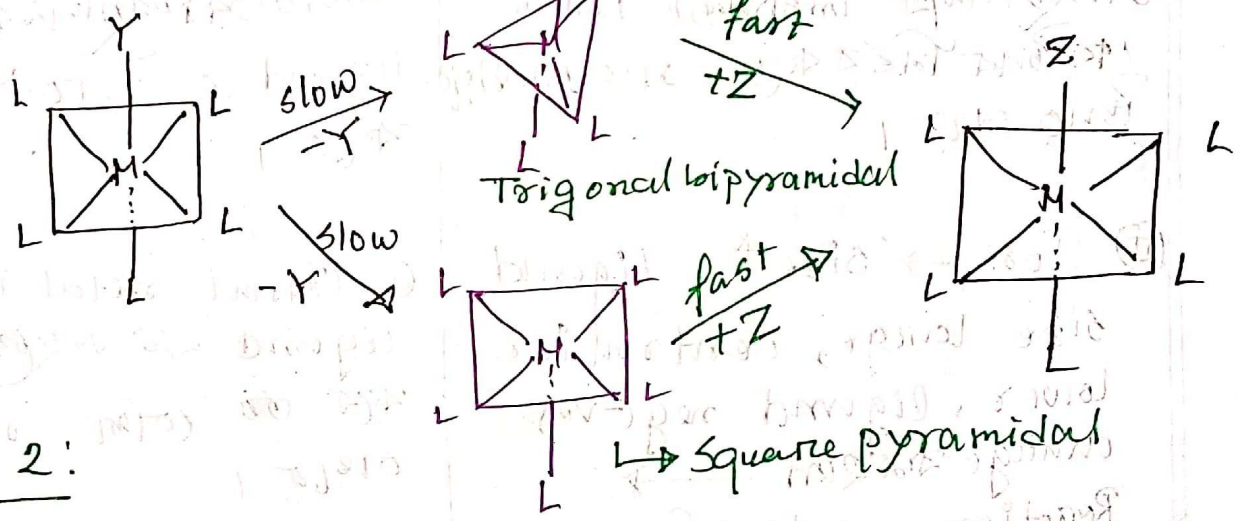
① S_N1 (dissociative) & (II) S_N2 (Associative)

Difinition

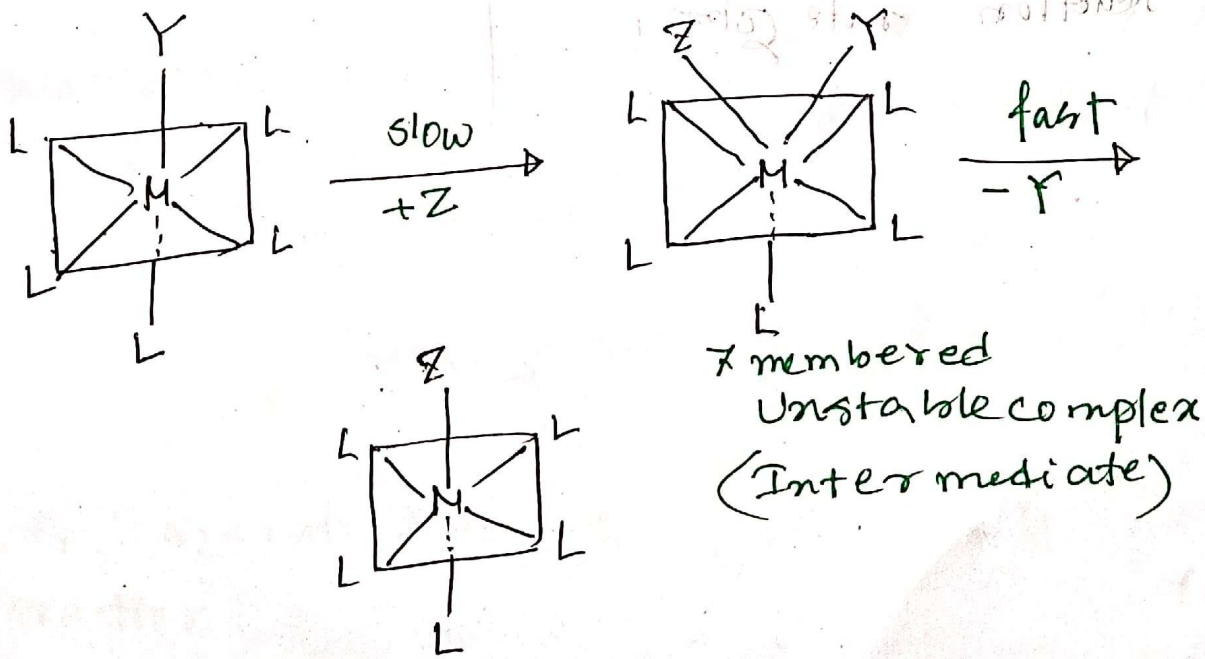
F-20

$$[L_5MY] + Z \longrightarrow [L_5MZ] + Y$$

F-20 S_N1 :



F-20 S_N2 :



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· Usually → Octahedral → S_N1

Tetra/square planer → S_N2
hedral

S_N1

① Central metal ion এবং প্রতিস্থাপক লিগান্ডের Nature এর উপর নির্ভর করে। প্রতিস্থাপকারীর উপর নয়।

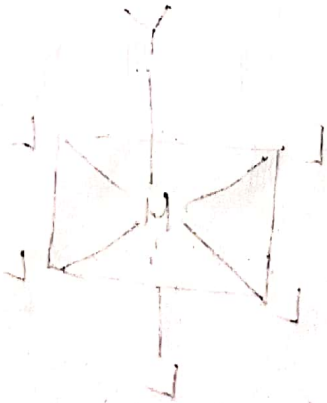
② Ion → size ↑, Ligand size large, central ion lower, ligand neg (-ve) charge বসালে → Reaction rate কমে।



S_N2

① Central ion এবং আসন্ন প্রতিস্থাপকারী Ligand-এর উপর নির্ভর করে।

② Central metal ion & Ligand এর মধ্যে bond বলে এর rate বাড়ে।



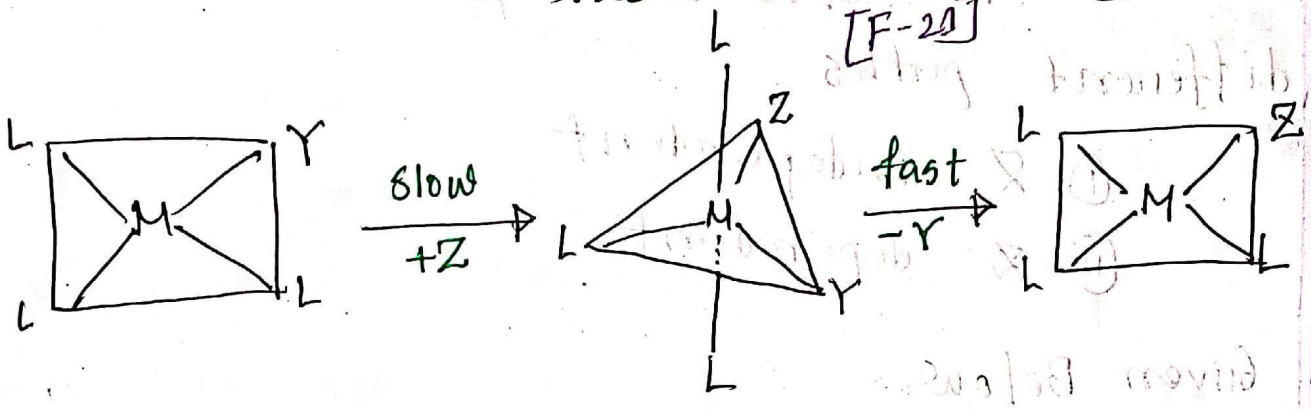


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Substitution of square planar Complex:

Discuss Mechanism (F-22) → (3)

[F-21]



2nd Order

$$ML_3Y + Z \longrightarrow ML_3Z + Y$$

$$\text{rate} = k[ML_3Y]^1[Z]^1$$

$$\text{Order} = 1+1 = 2$$

Water & other solvent may affect the rate of reaction. Because they can act as a ligand. So, the rate will be ~~as~~ -

$$-\frac{d}{dt}[ML_3Y] = K_1[ML_3Y] + K_2[ML_3Y][Z]$$

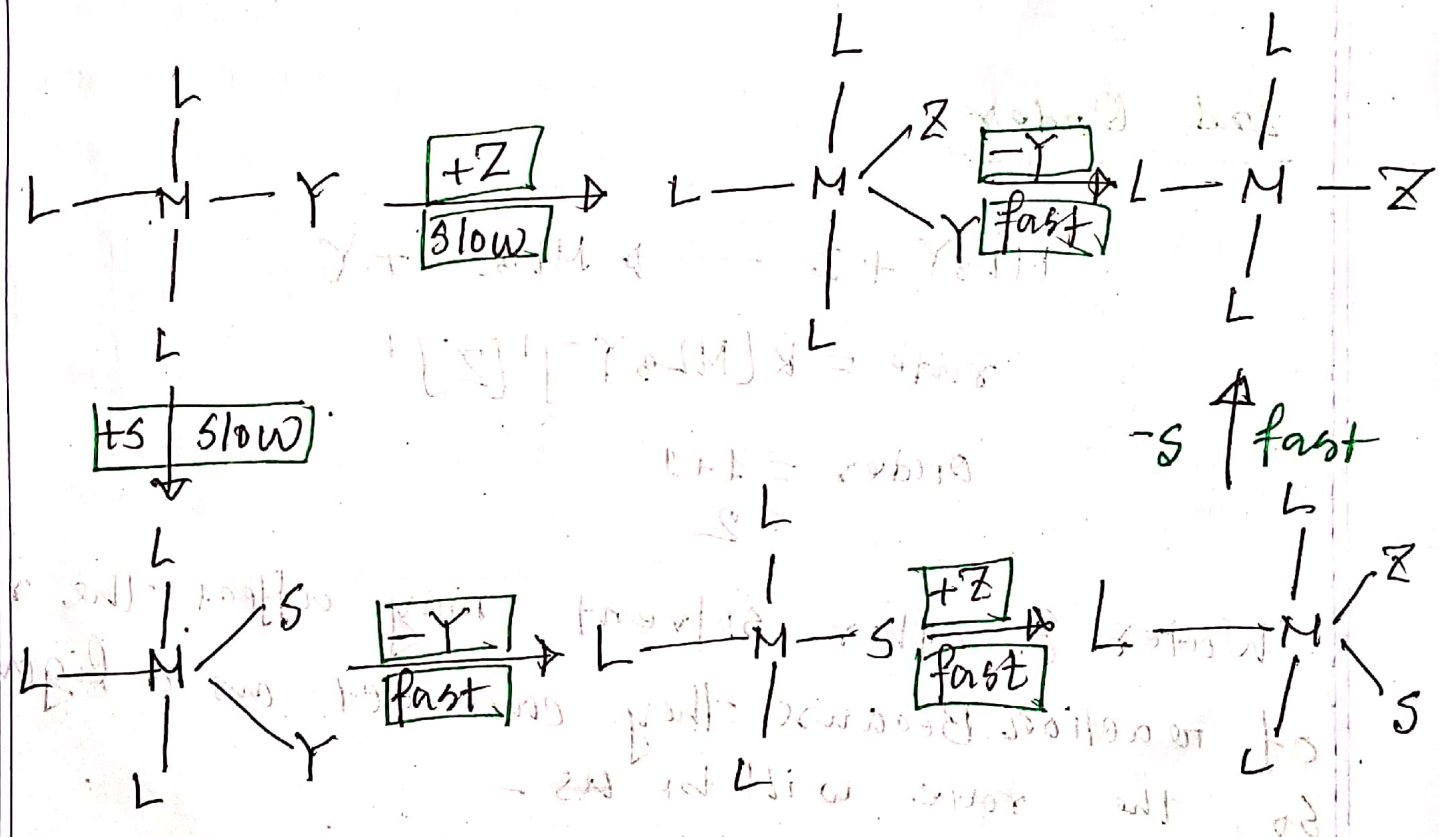
K_1 & K_2 are rate constants of 1st order & 2nd order reaction.

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From the two termed rate law, it is proved that, the reaction ~~with~~ proceeds in 2 different paths.

① Z : independent
② Z : dependent

Given Below:



Z Independent, S participate ~~no~~.

$$K_1 = K[S]$$

(H_2O, Alcohol)

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Z dependent. 2nd order, SN2 mechanism.

In solvent, CCl_4, C_6H_6, C_6H_{14} etc.

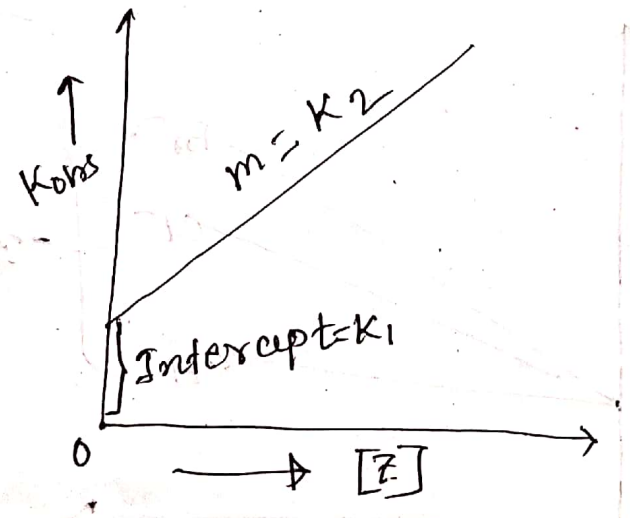
If the amount of ligands is very high then the reaction will be pseudofirst-order.

In this case,

$$-\frac{d}{dt}[ML_3Y] = K_1[ML_3Y] + K_2[ML_3Y][Z]$$
$$= \{K_1 + K_2[Z]\} \times [ML_3Y]$$
$$= K_{obs} \times [ML_3Y]$$

$$\therefore K_{obs} = K_1 + K_2[Z]$$

Plotting K_{obs} vs $[Z]$



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C_6H_{14}, CCl_4, C_6H_6 etc solvent এ $K_1 = 0$ হয়,

$$K_{obs} = K_2[Z]$$

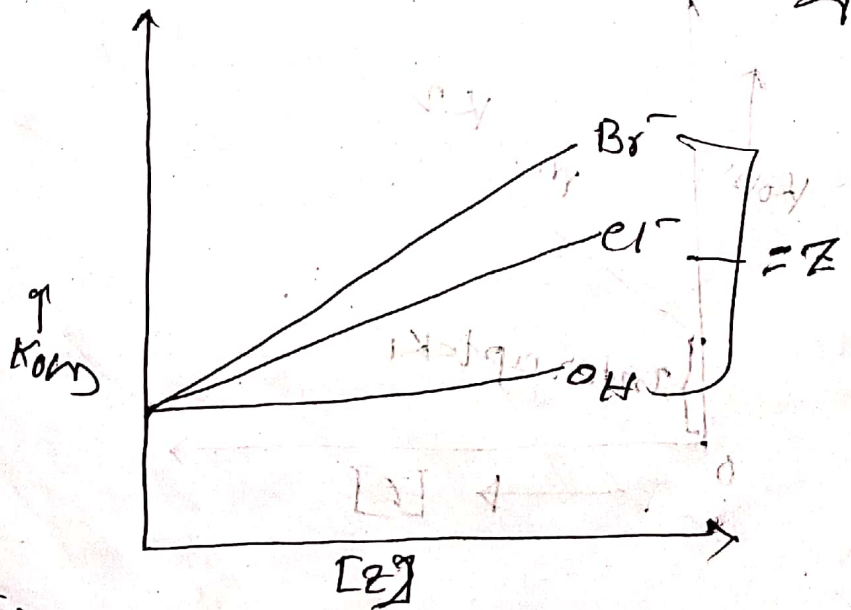
(ligand dependent)

In H_2O, C_2H_5OH etc দ্রাবক solvent path & ligand path এর মধ্যে competition হয়।

In a fixed solvent, different ligands react with complexes and getting K_{obs} vs $[Z]$ are straight line. These ~~intercept~~ intersect at the same point but the slopes are different.

K_1 constant but K_2 different

↳ be2 ligand different



$[Pt(dien)Cl]^+ + Z \rightarrow Pt[(en)Z]^{2+} + Cl^-$

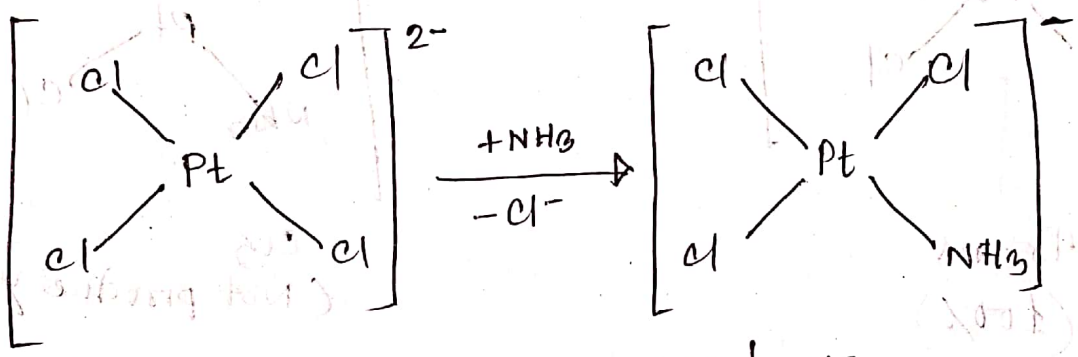
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Nucleophilicity sequence.

$$F^- < H_2O < NH_3 \sim Cl^- < Br^- < Py < I^- < CN^-$$

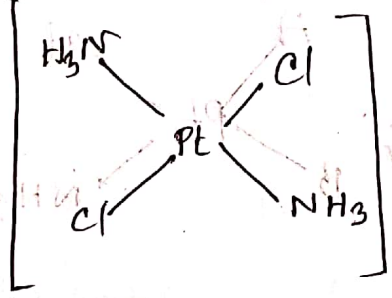
Trans effect: → Definition → F-22 [F-21]

Trans effect is the increase of lability of ligands that are trans to certain other ligands in the substitution reaction of square planar complex.



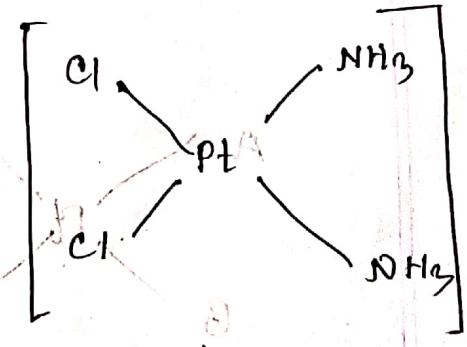
F-20 (products) → 4 marks

+NH3 | -Cl-



Trans isomer

(Not produce yet)



cis- isomer

(100%)

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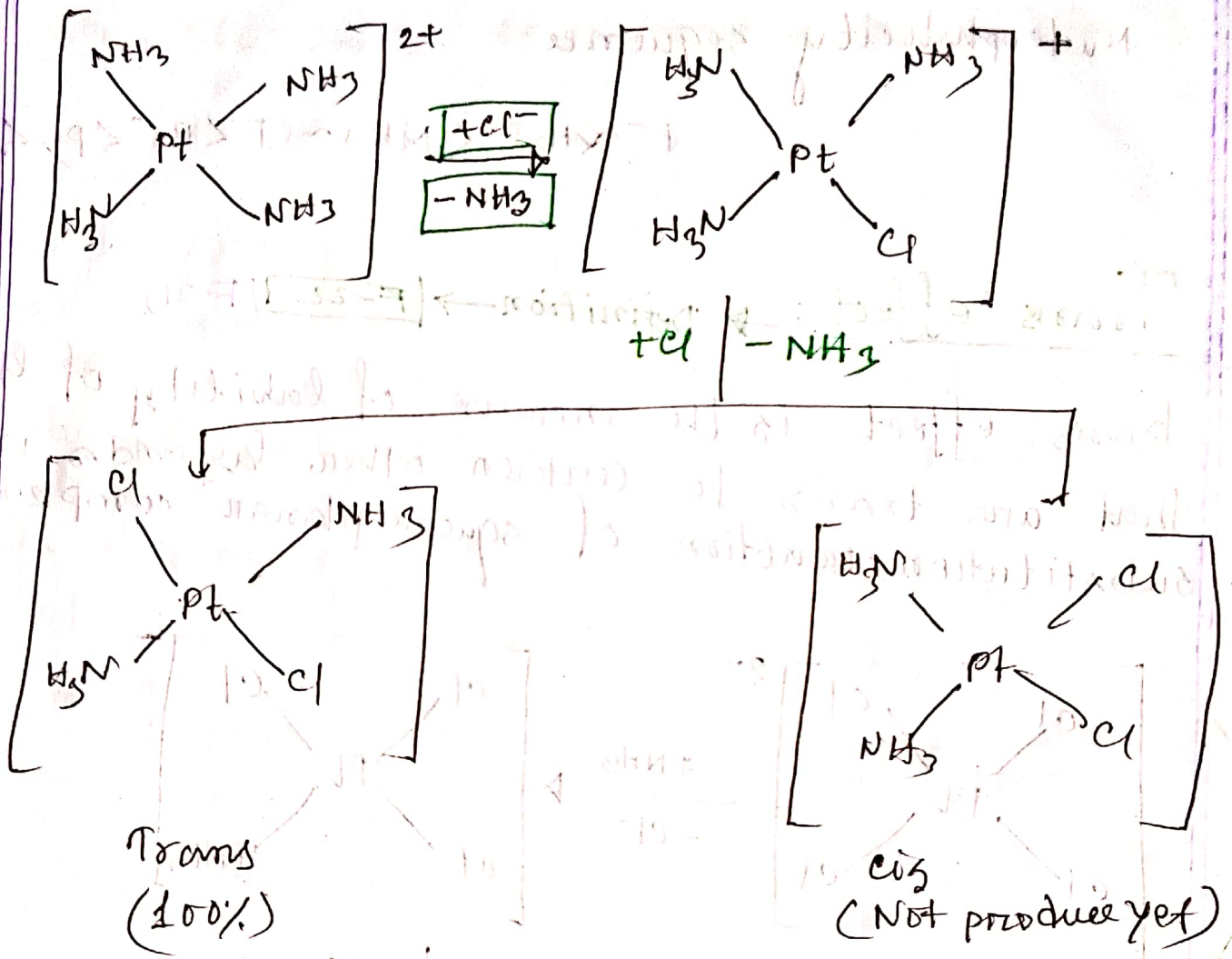
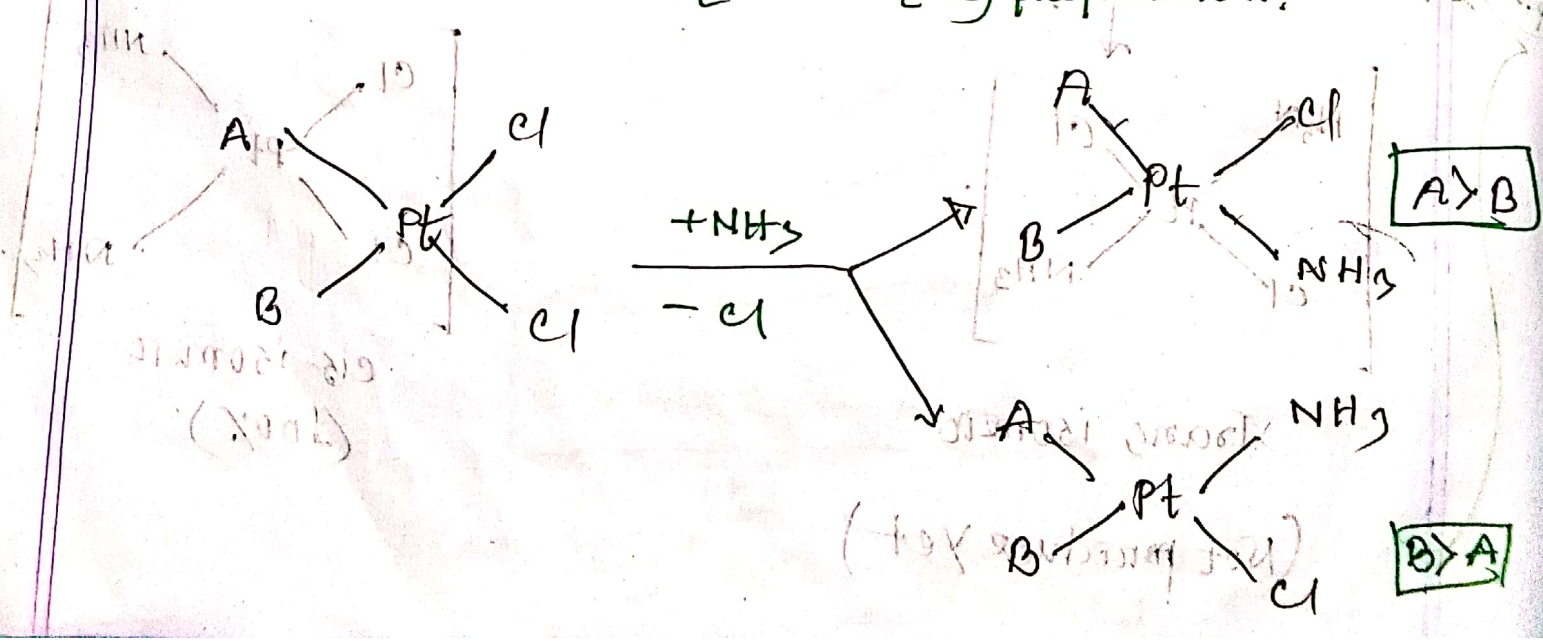


Fig: Two different methods of $[Pt(NH_3)_2Cl_2]$ preparation.



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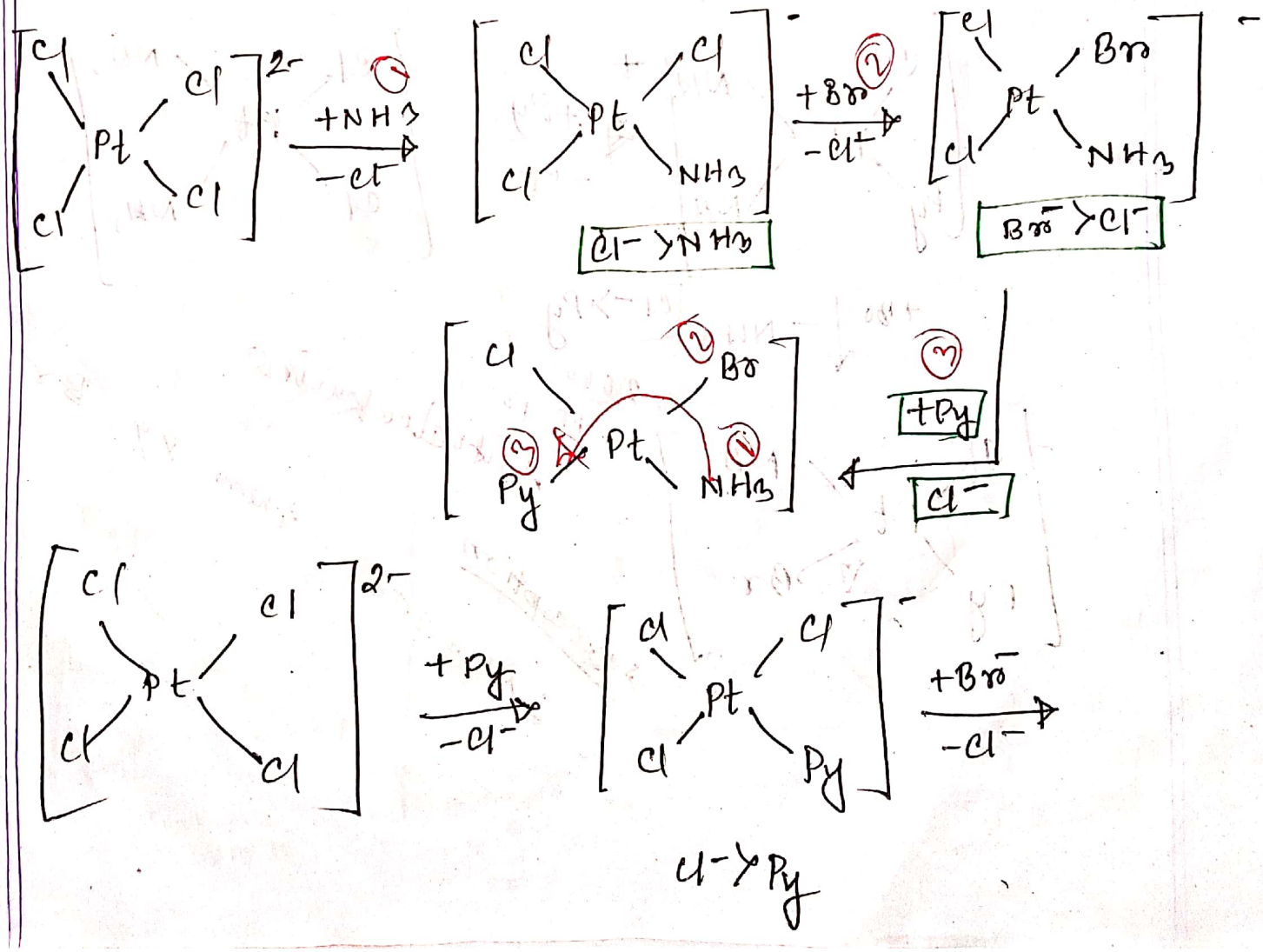
Sequence of trans effect power :

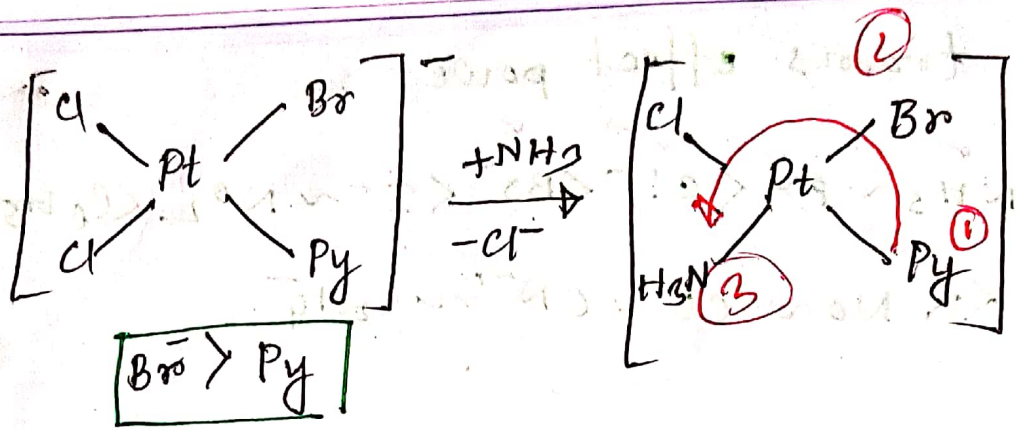
$H_2O \sim OH^- \sim NH_3 \sim Py < Cl^- < Br^- < I^- \sim NO_2^- < C_6H_5^- < CH_3^-$

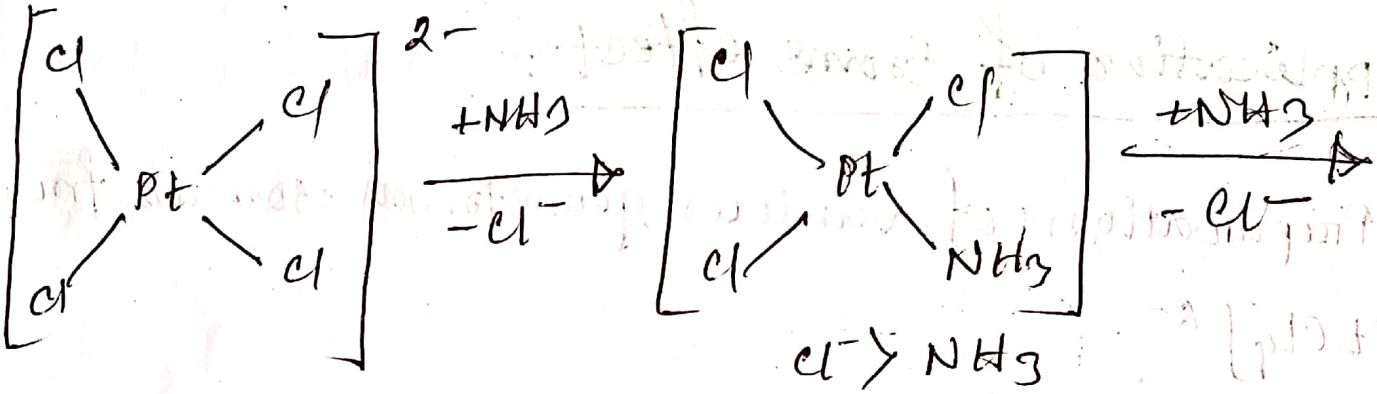
$< PR_3 \sim H^- << NO \sim CO \sim CN^- \sim C_2H_4$

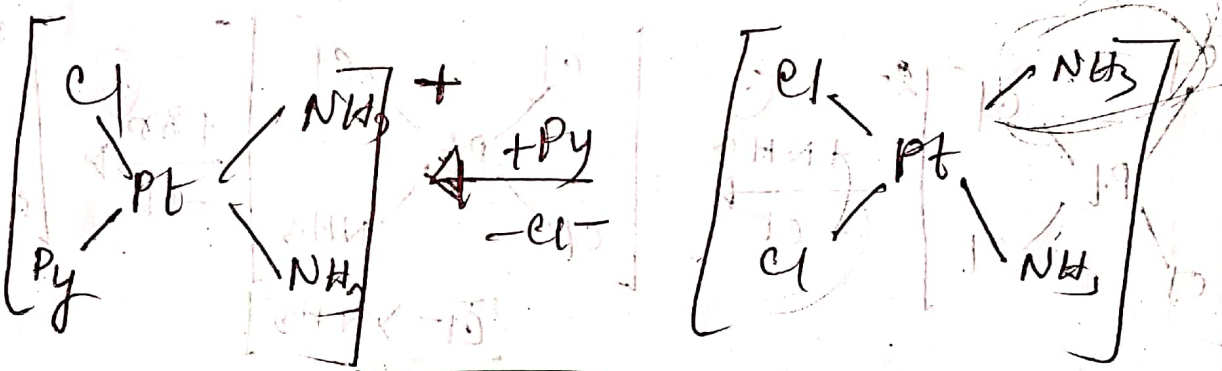
Application of trans effect :

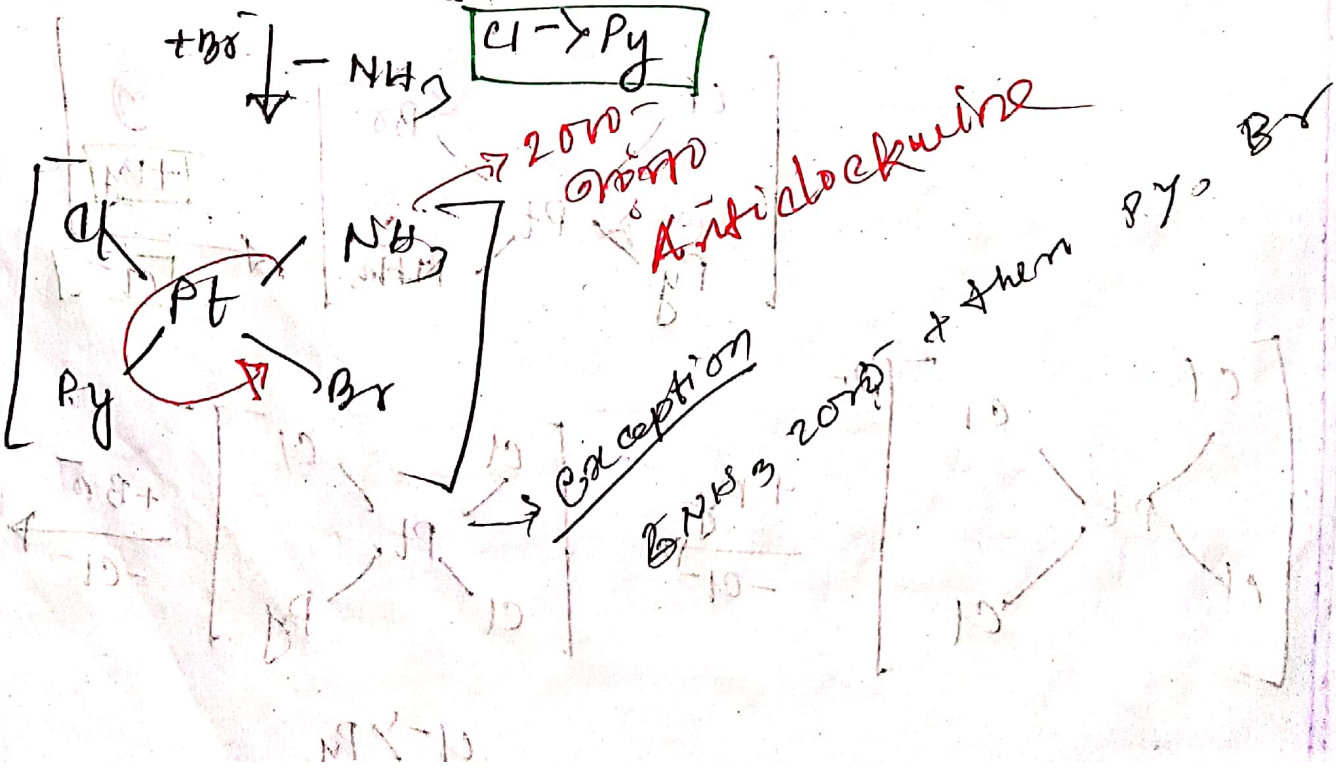
Preparation of various geometrical isomers from the $[PtCl_4]^{2-}$.









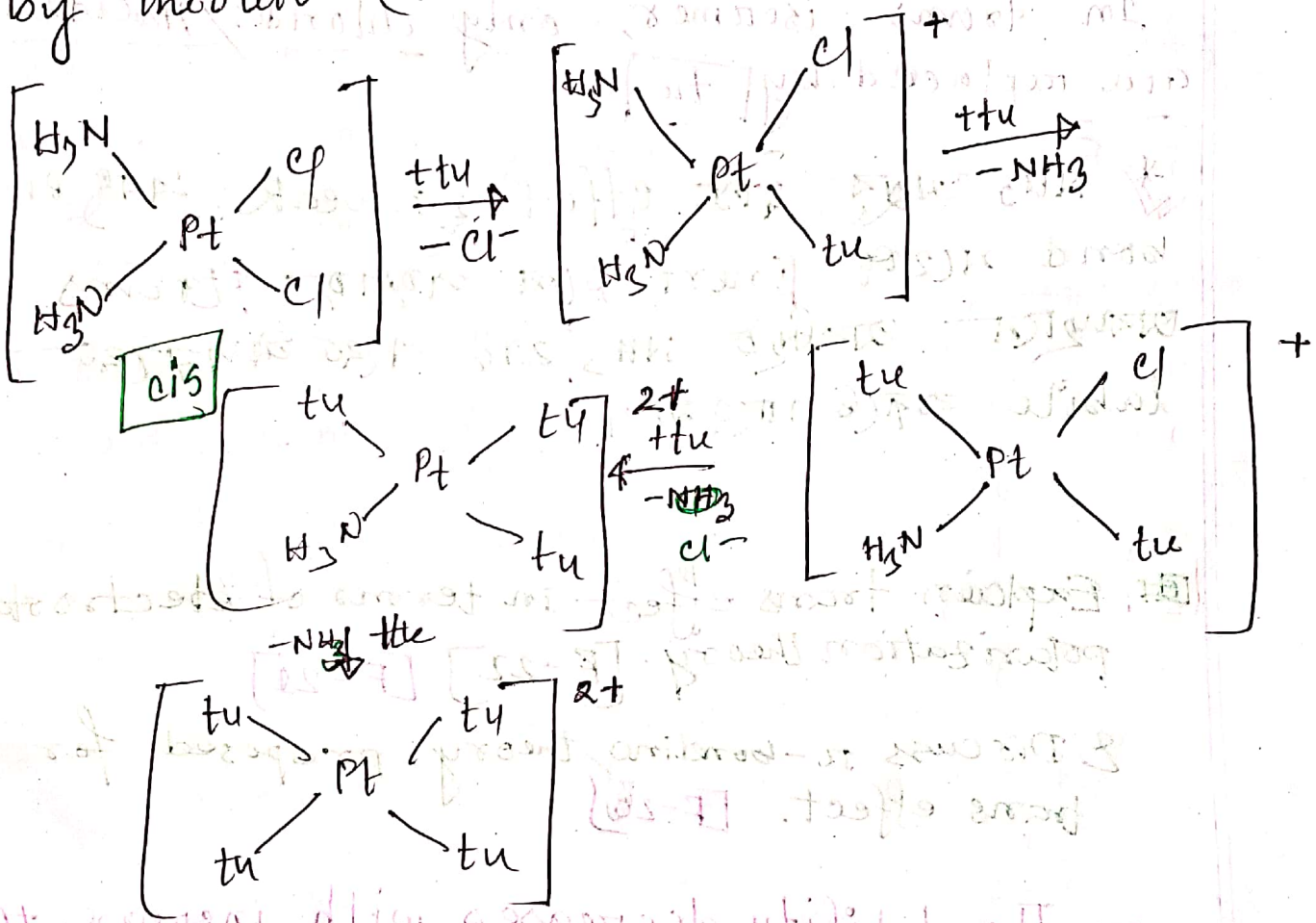


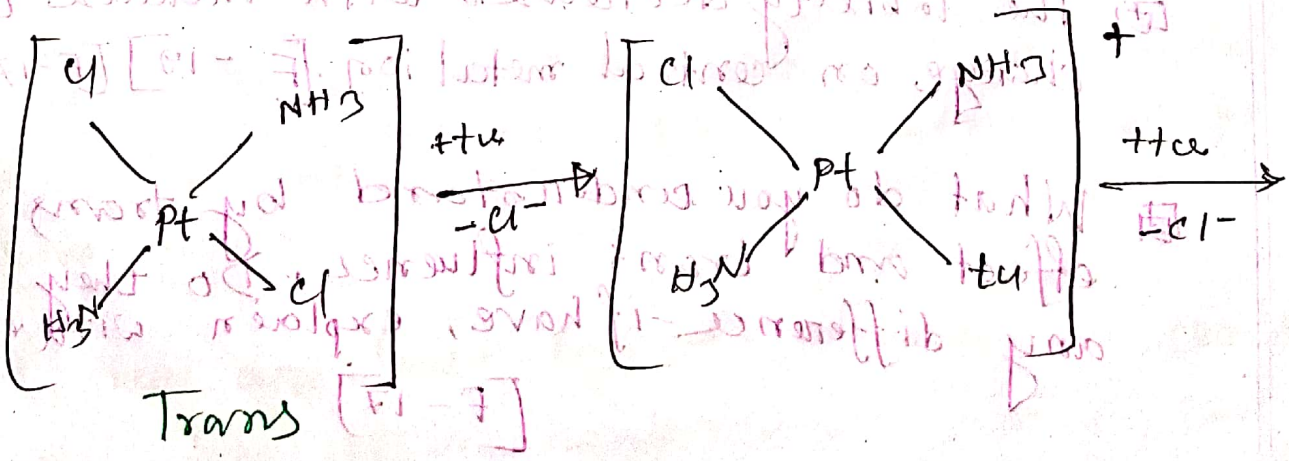
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A significant Application:

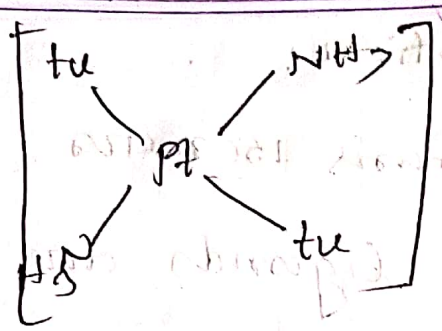
Differentiation of cis-trans isomers.

In cis isomers all the ligands are replaced by thiourea (tu)





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In trans isomer, only chloride/halides are replaced by [tu]

* NH_3 অণুর trans effect খুব weak. এবং Pt–N bond খুবই inert হলে সহজে trans অবস্থানে অবস্থিত NH_3 এর এই অপসারণকে labile করতে পারে না।

[F] Explain trans effect in terms of electrostatic polarization theory. [F-22] [F-21]

& Discuss π-bonding theory proposed for trans effect. [F-20]

[F] The lability decreases with increase the charge on central metal ion. [F-19] [F-17]

[F] What do you understand by trans effect and trans influence. Do they have any difference—if have, explain with example [F-17]

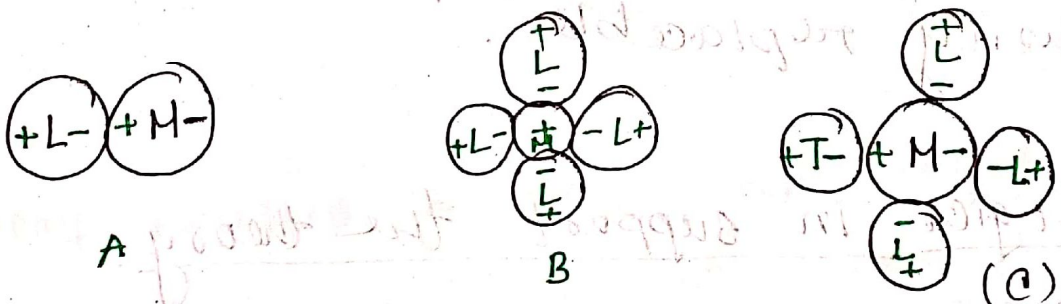
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* Explain reaction mechanism in trans and cis - $[M(LL)_2BX]$ (LL= bidentate such as en) [F-17]

* $[AlF_6]^{3-}$, $[SiF_6]^{2-}$, $[PF_6]^-$ and SF_6 are all six coordinate complexes. Arrange them in order of decreasing ligand exchange rate and explain (2+2) [F-16]

* What is the difference between intermediate and transition state (F-16) [2]

Electrostatic polarization Energy



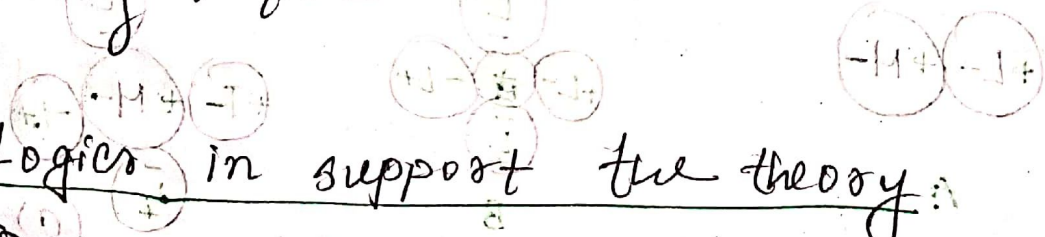
Due to the presence of ligand there created a dipole in metal ion which is drawn in the above figure

If there are four ligands around the central metal atom of the same types, then the

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the resultant of ligands opposite to each other are cancell out. So, net dipole = 0.

Let, one of the 4 ligands is more polarizable in this case, there creates a dipole in the metal ion. i.e. given in fig-(c). Here, same type of ligand's effect is cancel out and no charges creat on those sites. So, In this situation, the trans position of 'T' has the ligand 'L' where the positive charge is less on the metal ion. As a result, M-L bond will be weak and distance increas. Hence, L is easily replaceble.



Logics in support the theory:

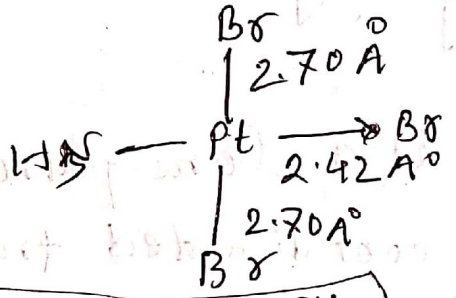
① The more the size of metal ion the more the polarization will be. As a result trans effect also increase.

$Pt^{(II)} > Pd^{(II)} > Ni^{(II)}$

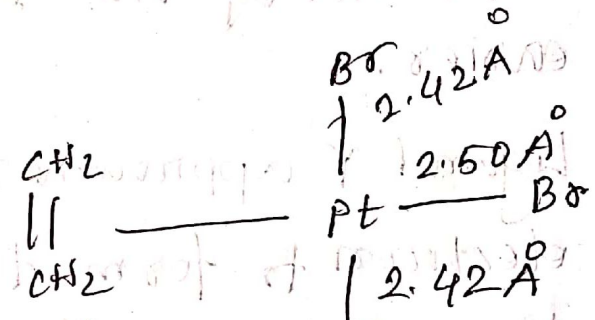
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(2) Ligand's size, double or triple bond ↑ more polarizable and trans effect ↑ বেশি হবে। Ex: $CN^-, CO, C_2H_4 > I^- > Br^- > Cl^-$

(3) $T-M-L$ (L, L), The trans position of of T is M-L. This bond will be weaker and distance increases.



T.E. : $Br > NH_3$



π-bonding Theory:

The theory explains the high trans effect shown by the π acceptor ligands like CO, NO, CN^- C_2H_4 etc. According to this theory the vacant π or π* orbitals of the π-acceptor ligand accepts electron pair from the metal to form pi-bond between M-L which stabilizes the transition state.

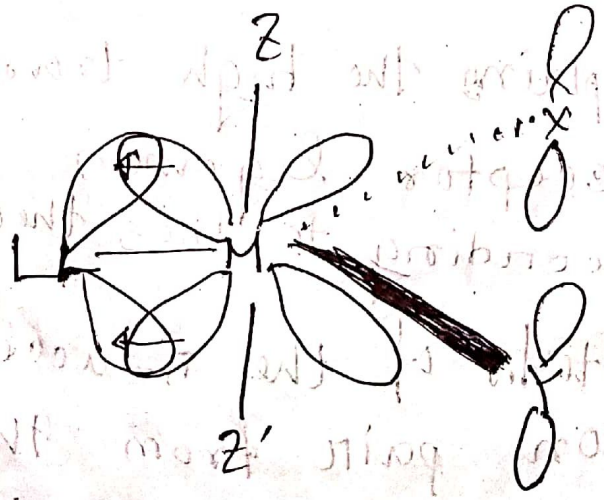
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The withdrawal of electron from metal by ligand increases the e- density toward M-L bond and diminishes in the direction of M-X.

L - M - X

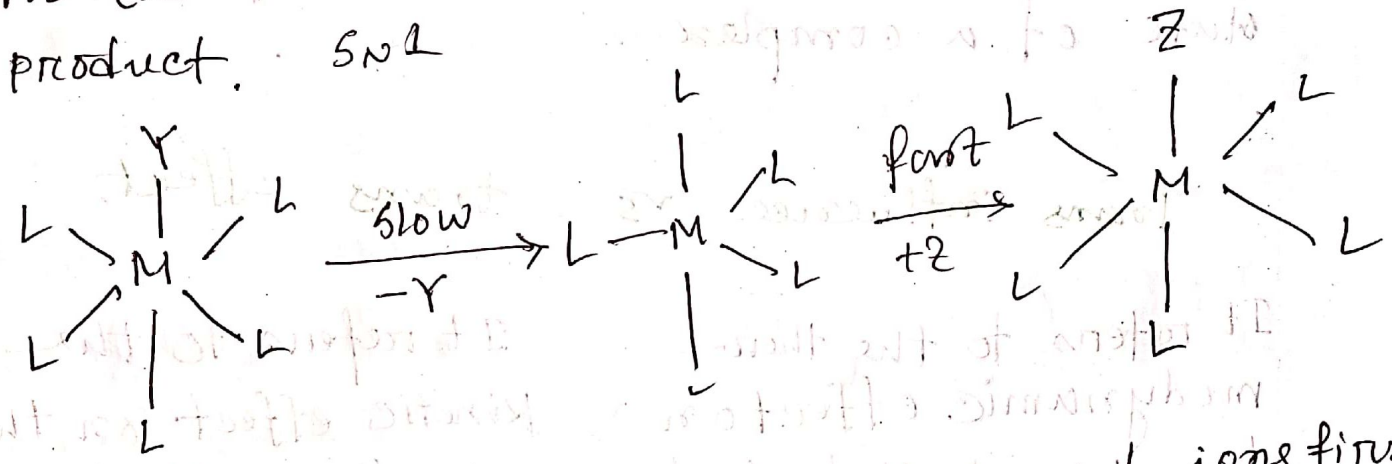
Thus the M-X bonds becomes weaker and entrance of an incoming ligand is become easier.

Ligand Y approaches with it's lone pair of electron to form the five coordinated transition state complex. L will stabilize the transition state by accepting the electron density that the incoming ligand Y donates to the metal and therefore facilitates substitution.

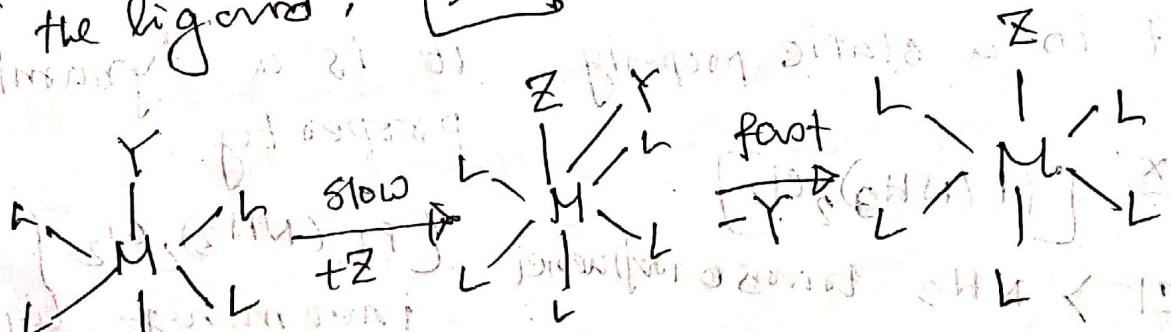


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In Dissociative Mechanism the metal ion dissociates the leaving ligand in the first step to form a five coordinate intermediate that is trigonal bipyramidal and then adds the incoming ligand in the second step to form the substituted product. SN1



In associative mechanism the metal ions first adds incoming ligand by accepting electron from the ligand and forms a seven coordinate compound and then dissociation occurs of the ligand. SN2



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Trans effect : ট্রান্স এফেক্ট

Trans influence: Trans influence refers to the impact of a ligand on the length of the bond trans to it in the ground state of a complex.

Trans influence vs	trans effect.
It refers to the thermodynamic effect on the ground state bond length, meaning how much a ligand weakens the bond trans to it	It refers to the kinetic effect on the rate of substitution of a ligand, meaning how much a ligand facilitates the replacement of ligand trans to it during reaction
It is a static property	it is a dynamic property.
Ex: $[Pt(NH_3)_2Cl_2]$ $Cl^- > NH_3$ trans influence meaning that Pt–NH_3 bond leng will be longer	$[Pt(NH_3)_2Cl_2]$ incoming ligand is more likely to replace the NH_3

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The high spin d^4 complex $[Cr(H_2O)_6]^{2+}$ is labile but low spin d^4 complex ion $[Cr(CN)_6]^{4-}$ is inert. Clarify the statement.

Soln:

A high spin d^4 complex like $[Cr(H_2O)_6]^{2+}$ is labile because water is a weak field ligand, resulting in a large no. of unpaired e^- in the d-orbitals, making the metal-ligand bonds weaker and more susceptible to substitution.

While a low spin d^4 complex like $[Cr(CN)_6]^{4-}$ is inert due to the strong field cyanide ligands which pair up the electrons, leading to a more stable complex with stronger metal-ligand bonds and less tendency to exchange ligands.

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Substitution rxn of Octahedral complex

Generally Octahedral complex follows the S_N1.

$$[M(H_2O)_6]^{n+} + L^{2-} \longrightarrow [M(H_2O)_5L]^{(n-2)+} + H_2O$$

$$\text{rate} = K[M(H_2O)_6]^{n+}$$

$$[M(H_2O)_6]^{n+} \xrightarrow[-H_2O]{\text{slow}} [M(H_2O)_5]^{n+} \xrightarrow[+L]{\text{fast}} [M(H_2O)_5L]^{(n-2)+}$$

O^{18} marked H_2O^{18} exchange with H_2O in the rxn, we observe that the more the charge of metal ion, the less the rate of rxn.

$$[M(H_2O)_6]^{n+} + 6H_2O^{18} \longrightarrow [M(H_2O^{18})_6]^{n+} + 6H_2O$$

$[Al(H_2O)_6]^{3+} < [Mg(H_2O)_6]^{2+} < [Na(H_2O)_6]^{+}$

size → less → rate of rxn low

$[Mg(H_2O)_6]^{2+} < [Ca(H_2O)_6]^{2+} < [Sr(H_2O)_6]^{2+}$

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Strong bond ⟶ slow rxn.

(18)

Maximum Octahedral complex lowspin follow

Eg: $[Co(NH_3)_5X]^{2+}$. $[X = Cl^-, Br^-, I^-, SCN^-]$

$$[Co(NH_3)_5X]^{2+} + H_2O \longrightarrow [Co(NH_3)_5(H_2O)]^{3+} + X^-$$

(Acidic Hydrolysis)

1st order. $K = K'[H_2O] = K' \times 55.55$

$$\text{rate} = K[Co(NH_3)_5X^{2+}]$$

$$\text{rate} = K'[Co(NH_3)_5X^{2+}][H_2O]$$

* In order to know the Mechanism: Method-I

If the conc. of e⁻ on metallic ion increases then the rate of rxn also increases. And the reaction proceeds easily if the metal-ligand bond is weak. Metal-Ligand (M-L) bond will be weak if the e⁻ donating ligand is added for increasing e⁻ density on metal. In this case, rxn rate increases. And different pyridines are added

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L-L

$NH_2CH_2CH_2NH_2$

$NH_2CH_2CH(CH_3)NH_2$

$NH_2C(CH_3)_2C(CH_3)_2NH_2$

rate

1.9×10^{-3}

3.7×10^{-3}

So fast.

In this case, if SN1 occurs, then one Cl- ~~left~~ leaves and converted in to five coordination number intermediate active compound. In this mechanism, more steric crowding more rxn rate. On the other hand, if crowding in SN2 increases, one extra ligand can form 7-membered intermediate which is formation is tough. So, rxn rate decreases. From the exp. we can see that the rxn rate increase as the increasing crowding. So, it follows SN1 rxn.

*

$$[Co(NH_3)_5X]^{2+} + Y^- \longrightarrow [Co(NH_3)_5Y]^{2+} + X^{\ominus}$$

Step-1:

$$[Co(NH_3)_5X]^{2+} + H_2O \xrightarrow[\text{RDS}]{\text{SLOW}} [Co(NH_3)_5(H_2O)]^{3+} + X^{\ominus}$$

Step-2:

$$[Co(NH_3)_5(H_2O)]^{3+} + Y^- \xrightarrow{\text{fast}} [Co(NH_3)_5Y]^{2+} + H_2O$$

Reaction of $[Co(NH_3)_5Cl]^{2+}$ complex with $\overset{\ominus}{O}H$ (Basic)

→ rate of rxn is 10^6 times of Acidic hydrolysis and it follows 2nd order.

$$[Co(NH_3)_5Cl]^{2+} + \bar{O}H \longrightarrow [Co(NH_3)_5(OH)]^{2+} + Cl^{\ominus}$$

$$\text{rate} = k_B[Co(NH_3)_5Cl]^{2+}[\bar{O}H]$$

Mechanism:

$$[Co(NH_3)_5Cl]^{2+} + \bar{O}H \xrightarrow{\text{SLOW}} [Co(NH_3)_5Cl(OH)]^{+}$$

$$\xrightarrow{\text{fast}} [Co(NH_3)_5OH]^{2+} + Cl^-$$

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But an alternative dissociation (S_N2) Mechanism is ——

$$[Co(NH_3)_5Cl]^{2+} + OH^- \xrightleftharpoons{\text{fast}} [Co(NH_3)_4(NH_2)Cl]^+ + H_2O \quad \text{---(1)}$$

$$[Co(NH_3)_4(NH_2)Cl]^+ \xrightarrow[K_2]{\text{slow}} [Co(NH_3)_4(NH_2)]^{2+} + Cl^- \quad \text{---(2)}$$

$$[Co(NH_3)_4(NH_2)]^{2+} + H_2O \xrightarrow{\text{fast}} [Co(NH_3)_5(OH)]^{2+} \quad \text{---(3)}$$

This reaction proceeds through the conjugate base $[Co(NH_3)_4(NH_2)Cl]^+$ of S_N1 of dissociation so, it is called S_N1CB.

From chemical equilibrium, From (1) ⇒

$$K = \frac{[Co(NH_3)_4(NH_2)Cl]^+ [H_2O]}{[Co(NH_3)_5Cl(OH)]^{2+} [OH^-]}$$

From (2) ⇒

$$\Rightarrow [Co(NH_3)_4(NH_2)Cl]^+ = \frac{K[Co(NH_3)_5Cl]^{2+}[OH^-]}{[H_2O]}$$

From (2) ⇒

$$\text{rate} = K_2[Co(NH_3)_4(NH_2)Cl]^+$$

$$= KK_2 \frac{[Co(NH_3)_5Cl]^{2+}[OH^-]}{[H_2O]}$$

$$= K_B[Co(NH_3)_5Cl]^{2+}[OH^-]$$

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$$K_B = \frac{K_2 K}{[H_2O]} = \frac{K_2 K}{55.5}$$ where,

SN1

But it has order (2).

Logics in support in SN1CB:

(I) Maximum ligands react with Co. order complex through SN1. so, with $\bar{O}H$, there is no logic to give SN2.

(II) Pt(II) complex gives SN2 with $\bar{O}H$ (weak nu) Co(III) → SN2 → there is no reason of more activity of $\bar{O}H$.

(III) The anions which act as strong nu but cannot remove H^+ ion from complex are not able to increase rate like $\bar{O}H$.

(IV) The complexes which have ionizable Hydrogens in ligands are showing SN1CB

Ex: $[Co(py)_4Cl_2]^+$ & $[Co(CN)_5Cl]^{3-}$

Not give SN1CB

(5) $\bar{O}H$ doesn't directly attack, instead but it removes H^+ and increases rate..

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Redox Rxn of Complex compounds:

SUZON-50

২ প্রকার। যথা-

① Electron exchange rxn:

e⁻ স্থানান্তরের কারণে কোনো রাসায়নিক পরিবর্তন লক্ষ করা যায় না। যেমন:

$[Fe(CN)_6]^{4-}$ and $[Fe(CN)_6]^{3-}$

Isotope & NMR এর সাহায্যে experiment করা হয়।

যেমন:

$$[Mn^{5}O_4]^{-} + [MnO_4]^{2-} \rightleftharpoons [Mn^{5}O_4]^{2-} + [MnO_4]^{-}$$

(2) Chemical change লক্ষ করা যায়। এদেরকে oxidation reduction বলা হয়।

Ex:

$$[Co(NH_3)_5Cl]^{2+} + [Cr(H_2O)_6]^{2+} + 5H_3O^{+} \longrightarrow [Co(H_2O)_6]^{2+} + [Cr(H_2O)_5Cl]^{2+} + 5NH_4^{+}$$

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For Redox, there are two mechanism:

(1) Outer-sphere redox rxn:

The mechanism in which the co-ordination sphere of the metal ions are not altered is called outer-sphere mechanism (or, electron-transfer mechanism)

For example:

$$[Fe^{(II)}(CN)_6]^{4-} + [Mo^{(III)}(CN)_6]^{3-} \longrightarrow [Fe^{(III)}(CN)_6]^{3-} + [Mo^{(II)}(CN)_6]^{4-}$$

Here, O.N. of metal ion is changed. If the complex is isotopic labelled, then the change of O.N is understood.

$$[Fe^{*}(H_2O)_6]^{3+} + [Fe(H_2O)_6]^{2+} \longrightarrow [Fe^{*}(H_2O)_6]^{2+} + [Fe(H_2O)_6]^{3+}$$

$$[Fe^{II}(CN)_6]^{4-} + [Fe^{III}(CN)_6]^{3-} \longrightarrow [Fe^{III}(CN)_6] + [Fe^{II}(CN)_6]^{4-}$$

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Explanation:

In outer-sphere mechanism, the no. of ligand is not changed in before and after the rxn. Here, substitution rxn is not occure, so it is labile for substitution rxn.

Condition:

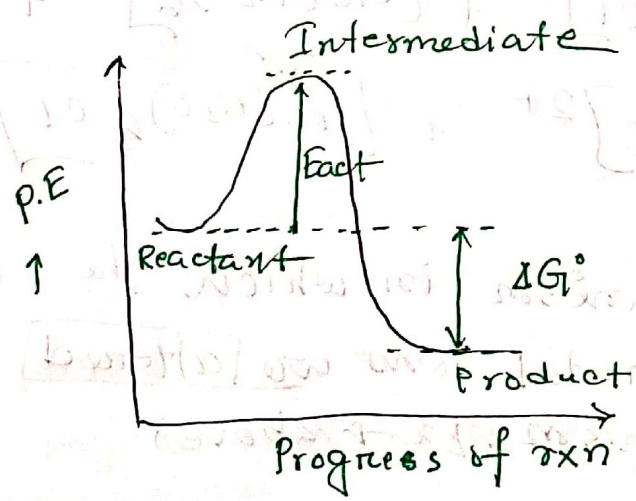
(I) The position of the ligand is not changed.

(II) The P.E of reactant and product are same

(III) Activation energy are same.

Ex: $Fe^{II} - O$, Fe^{2+} (Low spin) $\Rightarrow$ 75 pm

$Fe^{III} - O$, Fe^{3+} (Low spin) $\Rightarrow$ 69 pm



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* Atom or Group transfer of Inner-sphere mech

These are electron transfer rxn in which concomitant electrons transfer and ligand transfer occurs. In such rxn, an intima contact between oxidant and reductant is required. This intimate contacts is fulfilled when a bridged activated complex has at least one ligand which is common to the coordinate sphere of both the reacting comple and thin form a bridge betn them. Hence the name inner mechanism

For example:

In acidic medium,

$$[Co^{III}(NH_3)_5Cl]^{2+} + [Cr(H_2O)_6]^{2+} + H_3O^+ \longrightarrow$$

$$[Co^{II}(H_2O)_6]^{2+} + [Cr^{(III)}(H_2O)_5Cl]^{2+} + NH_4^+$$

The mechanism in which the co-ordination sphere of the metal ions are altered is called inner-sphere mechanism. Ex→ (above)

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Tu-R

Different steps of this: (Mechanism)

$$[ML_x]^{m+} + [NY_y]^{n+} \xrightarrow{-Y} L_{x-1}M - L - N\,Y_{y-1}$$

(e⁻ transfer)

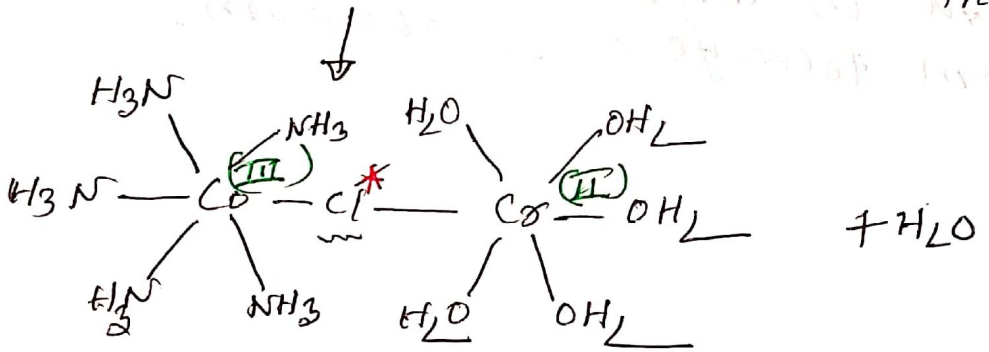
$$L_{x-1}M^{(m-1)+} \quad L \quad N^{(n+1)+}\,Y_{y-1}$$

↓

Products

Example:

$$[Cr(H_2O)_6]^{2+} + [Co(NH_3)_5Cl]^{2+}$$



Evidence:

When this reaction occurs, the Cl^- acts a bridging ligand, transferring from cobalt to Cr-complex during e⁻ transfer. By using radioactive Cl^- ions, researchers have shown that the transferred e⁻ primarily originates from Co-complex and supporting the inner-sphere mechanism.

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* For established this mechanism scientist did experiment with F^-, Br^-, I^- SO_4^{2-}, NCS^- instead of Cl^-. They saw the same result, In each case, the negative ion attached with $Cr(III)$. While the e⁻ transfer, the negative ion also transfer from oxidant to reductant. This mechanism is also called ligand transfer or atom transfer mechanism.

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